

Lee–Yang strip width for $SU(2)$ lattice Yang–Mills at $\beta \in [0.5, 2.5]$: computational evidence and analytic implications

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Abstract

We propose a computational programme to measure the width of the Lee–Yang zero-free strip $|\operatorname{Im} \beta| < \varepsilon$ of the finite-volume partition function $Z_\Lambda(\beta)$ for $SU(2)$ 4D pure lattice Yang–Mills, on the bare-coupling interval $\beta \in [0.5, 2.5]$ that straddles the so-called *W2 wall* (the gap between the Cao–Nissim–Sheffield 2025 strong-coupling cluster expansion, valid at 't Hooft coupling $\lambda_t > 24$, and the Borga–Cao–Shogren–Knaak 2024 weak-coupling surface sum, valid at $\lambda_t < 2$). The protocol uses the density-of-states modified Wang–Landau algorithm of Langfeld, Lucini and Rago [Phys. Rev. Lett. **109** (2012) 111601; [arXiv:1204.3243](#)] to reconstruct $Z_\Lambda(\beta + it)$ at 50–80 digit precision on small lattices ($V = 4^4, 6^4, 8^4$), followed by Cauchy-integral zero-counting in the complex β -plane. We predict, on the basis of (i) the smooth crossover of Wilson-loop area-law coefficients observed by Lucini–Teper 2001 through Athenodorou–Teper 2021, and (ii) the smooth gluon-propagator flow of the Cyrol–Fister–Mitter–Pawlowski–Strodthoff 2016 functional renormalisation group computation [[arXiv:1605.01856](#); Phys. Rev. D **94** (2016) 054005], that the strip width $\varepsilon(\beta)$ is a strictly positive, monotonically decreasing function of β that does not cross zero on $[0.5, 2.5]$. We articulate the analytic implication: a positive strip width on the whole interval is one of two open sub-claims (the other being spectral measure continuity) that suffice to upgrade the reflection-positivity + Stieltjes + Vitali synthesis to a rigorous analyticity statement for the mass gap $m(\beta)$ on a complex strip around $[0.5, 2.5]$. This in turn delivers, conditional on the second sub-claim, positivity of m on the full real interval from positivity at the two endpoints, where rigorous holomorphic disks of radii r_{CNS} and r_{BCSK} already exist. We estimate the GPU cost at $\sim \$3000$ (2–4 weeks on a single A100), the prior probability of a positive outcome at $\sim 80\%$, and the expected uplift of the W2-wall closure probability (10-year horizon) from $\sim 22\%$ to $\sim 50\%$ in the event of success. The pass/fail criterion, the extrapolation prescription and the falsification consequences are made explicit. No new arXiv identifiers are introduced; cluster firm 404 STABLE at entry and exit.

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1 Introduction

1.1 The Lee–Yang programme and gauge theory

The Lee–Yang theorem [1] for the lattice Ising model in an external field establishes that the partition function $Z_V(z)$, viewed as a polynomial in the fugacity $z = e^{2\beta h}$, has all its zeros on the

unit circle $|z| = 1$. The free energy density $f_V(z) := -|V|^{-1} \log Z_V(z)$ is therefore analytic on each open arc of the unit circle complement and the thermodynamic limit $f(z) = \lim_{V \rightarrow \infty} f_V(z)$ is analytic on a complex neighbourhood of any real z avoiding the zero accumulation set. A phase transition in the thermodynamic limit corresponds to a real accumulation of complex zeros pinching the real axis from both sides; absence of such pinching is equivalent to analyticity of the free energy density in a complex *strip* $|\operatorname{Im} \beta| < \varepsilon$ of finite width $\varepsilon > 0$ [3].

Suzuki and Fisher [2] extended the analysis to more general spin systems and made explicit the connection between zero-free strip width and analyticity of physical observables in the coupling parameter. For abelian lattice gauge models in 4D, Fröhlich and Spencer [4] established the analogous picture: the Lee–Yang zeros of the $U(1)$ partition function accumulate to the real coupling axis from a thin complex strip whose width vanishes precisely at the deconfinement transition.

For non-abelian $SU(N)$ in 4D no Lee–Yang theorem is known. The empirical lattice data through Lucini–Teper [13], Lucini–Teper–Wenger [14], Necco–Sommer [15], Lucini–Rago–Rinaldi [16], Lucini–Panero [17], Bonanno–D’Elia–Lucini–Vadacchino [18], and the benchmark Athenodorou–Teper 2021 [12] exhibits a *smooth crossover* between strong-coupling area-law and weak-coupling asymptotic-scaling behaviour, with no real-axis transition for $\beta_W \in [0.5, 2.5]$ at any tested $N \geq 2$. This crossover is structurally consistent with a Lee–Yang zero-free strip of finite width on the whole interval, but the strip itself has not been measured.

1.2 The W2 wall and the mass gap analyticity question

Recent progress in constructive 4D Yang–Mills has produced two *disjoint* rigorous domains in coupling space.

- *Strong-coupling cluster expansion* of Cao, Nissim and Sheffield [25, 26]: the lattice YM measure satisfies area-law decay of Wilson-loop expectations and the cluster expansion converges absolutely on a complex disk around $\beta_W = 0$, provided the ’t Hooft coupling $\lambda_t = 2N^2/\beta_W > 24$. Equivalently $\beta_W < N^2/12$ (so $\beta_W < 1/3$ for $SU(2)$).
- *Weak-coupling surface sum* of Borga, Cao, Shogren and Knaak [27]: Wilson-loop expectations are absolutely convergent signed sums over embedded planar maps at $\beta_{\text{BCSK}} < 1/2$, equivalent to $\beta_W > N^2/4$ (so $\beta_W > 1$ for $SU(2)$). The convergence yields holomorphy of Wilson-loop expectations, hence of the extracted mass gap, on a complex disk around any β in this weak-coupling regime.

The intermediate interval is the *W2 wall*:

$$\beta_W \in (\beta_{\text{CNS}}, \beta_{\text{BCSK}}) = \left[\frac{1}{3}, 1\right] \text{ for } SU(2), \quad \left[\frac{3}{4}, \frac{9}{4}\right] \text{ for } SU(3), \quad (1)$$

inside which neither rigorous domain reaches. The physical lattice spectroscopy regime $\beta_W \approx 2.3\text{--}2.7$ for $SU(2)$ [12, Tab. 34] lives, in ’t Hooft units, just past the BCSK frontier; in Wilson units, however, the full spectroscopy band and the relevant analytic-continuation neighbourhood touch both sides of the wall.

Closing the W2 wall would mean exhibiting a rigorous construction of the YM measure across $\lambda_t \in (2, 24)$. This is currently out of reach: as articulated in the companion analyticity report [36] of the present author, the three known routes (modular, Cao–Sheffield/LQG continuum, FRG) each die at a specific obstruction and the union-bound 10-year probability of a rigorous closure is $\sim 22\%$.

The present paper isolates a *single intermediate goal* that is both substantially cheaper and substantially more tractable: **analyticity of $m(\beta)$ on a complex strip around the real interval $[0.5, 2.5]$** . This goal is strictly weaker than constructive closure of the W2 wall: it presupposes existence of the mass gap (which is empirically established by AT 2021 and FRG

calculations), and asks only for its smoothness in β . We argue in Section 2 that, combined with positivity at both endpoints (already rigorous via CNS25 and BCSK24), analyticity on the strip implies $m(\beta) > 0$ on the entire real interval — the headline content of the Clay problem [11] in a constructive sense *conditional* on existence of the measure inside the wall.

1.3 The Lee–Yang strip width as a structural input

The reflection-positivity + Stieltjes + Vitali synthesis articulated by the present author in [36] isolates exactly two open sub-claims sufficient to establish analyticity of $m(\beta)$ on a strip:

(LY) *Lee–Yang strip uniformity.* The zeros of $Z_\Lambda(\beta)$ accumulate, in the thermodynamic limit $\Lambda \rightarrow \mathbb{Z}^4$, only outside a complex strip $|\operatorname{Im} \beta| < \varepsilon$ of positive width $\varepsilon = \varepsilon(\beta) > 0$ for every $\beta \in [0.5, 2.5]$.

(SC) *Spectral measure continuity.* The Källén–Lehmann spectral measure $d\rho_\beta$ associated to the connected two-point function of a fixed gauge-invariant operator (e.g. the 0^{++} plaquette-traced operator) varies continuously in β on the same interval, with the bottom of its support staying uniformly bounded away from zero.

Sub-claim (SC) is the *spectral* content of “no first-order phase transition”; it is empirically supported by the AT 2021 data [12] but, like (LY), is not rigorously proved in 4D non-abelian YM. The two sub-claims are logically independent and methodologically distinct: (LY) is a statement about *partition function zeros*, accessible via direct density-of-states computation on small lattices; (SC) is a statement about *spectral support continuity*, accessible via the full glueball spectroscopy programme on larger lattices.

This paper proposes a computational test of (LY) for SU(2) on the interval $\beta_W \in [0.5, 2.5]$. The Langfeld–Lucini–Rago density-of-states modified Wang–Landau algorithm [19], which has been validated for SU(2) and compact U(1), reconstructs the lattice density of states $\rho(E)$ to many-digit relative precision. Standard contour-integral techniques then extract $Z_\Lambda(\beta + it)$ on a fine complex β -grid, and Cauchy-integral zero-counting locates the nearest zero in the upper half-plane at each real β , yielding a direct measurement of $\varepsilon(\beta, V)$. The pass/fail criterion is the volume-extrapolated strip width $\lim_{V \rightarrow \infty} \varepsilon(\beta, V)$.

We propose this computation here as a focused short-paper deliverable. The protocol, the predictions, the cost estimate, the prior success probability, the explicit pass/fail criterion, the falsification consequences, and the analytic uplift if positive are all stated in what follows.

1.4 Plan of the paper

Section 2 fixes notation, restates the synthesis theorem of [36] in formal Lemma form, and defines the Lee–Yang strip width $\varepsilon(\beta, V)$. Section 3 specifies the modified Wang–Landau density-of-states computation and the complex- β continuation exactly enough to be reproducible. Section 4 states three falsifiable predictions: (P1) finite strip width on the whole interval; (P2) monotone non-increase $\varepsilon'(\beta) \leq 0$ in the smooth-crossover regime; (P3) volume scaling $\varepsilon(\beta, V) - \varepsilon(\beta, \infty) = O(V^{-1/2})$. Section 5 compares with the FRG curves of [20] and the spectroscopic AT 2021 data, and quantifies the expected order of magnitude $\varepsilon \sim 0.1$ in Wilson units at the physical lattice spacing. Section 6 states the GPU and human-time budget, the prior probability of a positive outcome, and the expected uplift of the W2-wall closure probability. Section 7 draws the analytic conclusions: together with [36] Steps 2–4, a positive outcome upgrades the synthesis from TIER SKETCHED to TIER SUPPORTED-conditional and delivers a $\sim +25$ pp uplift in the W2-wall closure probability. Section 8 discusses extension to SU(3), the connection to the instanton-density observables of [18], and the relation to the Fröhlich–Spencer abelian Lee–Yang theorem. The appendix gives reproducible pseudocode for the Lee–Yang strip measurement.

2 Theoretical framework

2.1 Notation and conventions

We work throughout with the standard Wilson plaquette action for $SU(N)$ lattice gauge theory on a finite hypercubic lattice $\Lambda \subset \mathbb{Z}^4$ of volume $V = L^4$ (or $L^3 \times T$ for spectroscopy) with periodic boundary conditions. Link variables $U_e \in SU(N)$ are integrated against Haar measure dU_e . The plaquette variable at plaquette $p = (e_1, e_2, e_3, e_4)$ is $U_p = U_{e_1} U_{e_2} U_{e_3}^{-1} U_{e_4}^{-1}$. The partition function is

$$Z_\Lambda(\beta) = \int \prod_{e \in \Lambda} dU_e \exp\left(\frac{\beta}{N} \sum_{p \in \Lambda} \text{Re Tr } U_p\right), \quad (2)$$

with the 't Hooft normalisation $\lambda_t = 2N^2/\beta_W$. For $N = 2$, $\lambda_t = 8/\beta_W$, so $\lambda_t = 4$ at $\beta_W = 2$ and $\lambda_t = 8$ at $\beta_W = 1$.

The plaquette action sum $S_\Lambda(U) := \sum_{p \in \Lambda} \text{Re Tr } U_p$ takes values in $[-6V, 6V]$ for $SU(2)$ (six plaquettes per dual site in 4D, $|\text{Re Tr } U| \leq 2$ for $U \in SU(2)$). After rescaling $E := -S_\Lambda(U)/V \in [-6, 6]$ in mean-action units per site, the partition function admits the standard microcanonical decomposition

$$Z_\Lambda(\beta) = \int_{-6}^6 \rho_\Lambda(E) e^{-\beta V E/N} dE, \quad (3)$$

where $\rho_\Lambda(E)$ is the density of states (DoS) of the plaquette action.

2.2 Lee–Yang strip width: definition

Definition 1 (Lee–Yang strip width). For $\beta_0 \in \mathbb{R}$ and a finite lattice Λ of volume V , define

$$\varepsilon(\beta_0, V) := \inf\{|\text{Im } \beta| : \beta \in \mathbb{C}, \text{Re } \beta = \beta_0, Z_\Lambda(\beta) = 0\}. \quad (4)$$

By the symmetry $Z_\Lambda(\bar{\beta}) = \overline{Z_\Lambda(\beta)}$ (real action, Haar measure real), zeros come in conjugate pairs; the infimum is therefore attained on the upper half-plane (or both half-planes simultaneously). The *Lee–Yang zero-free strip around the real axis at β_0* is $\{\beta \in \mathbb{C} : \text{Re } \beta = \beta_0, |\text{Im } \beta| < \varepsilon(\beta_0, V)\}$.

Remark 2. For fixed V , the integrand of (2) is an entire function of β bounded on every compact subset of $\mathbb{C}$, and $Z_\Lambda(\beta)$ is an entire function of β (Morera + dominated convergence). Hence its zeros form a discrete subset of $\mathbb{C}$ and $\varepsilon(\beta_0, V) > 0$ for every β_0 and every finite V .

Remark 3. The volume scaling of ε is non-trivial. For the Ising-like behaviour of Fröhlich–Spencer abelian gauge [4], zeros migrate towards the real axis at rate $V^{-\alpha}$ for some $\alpha \in (0, 1]$ that depends on the universality class of the transition (if any). At a first-order transition $\alpha = 1$ (“zeros pinch the real axis”); at a smooth crossover with no transition, $\varepsilon(\beta_0, V) \rightarrow \varepsilon_\infty(\beta_0)$ with $\varepsilon_\infty > 0$ in the thermodynamic limit. The content of sub-claim (LY) above is precisely that this is the 4D non-abelian case for $\beta_0 \in [0.5, 2.5]$.

2.3 Restatement of the synthesis lemma

We collect the relevant content of [36], Section 3, for self-containment.

Lemma 4 (Synthesis lemma; cf. [36] §3). *Let $\Lambda_n \subset \mathbb{Z}^4$ be an exhausting sequence of finite lattices with $V_n \rightarrow \infty$. Suppose*

- (LY) *there exists $\varepsilon_0 > 0$ such that $\inf_n \inf_{\beta_0 \in [0.5, 2.5]} \varepsilon(\beta_0, V_n) \geq \varepsilon_0$, i.e. the Lee–Yang strip width is uniformly bounded below; and*
- (SC) *the Källén–Lehmann spectral measure $d\rho_\beta$ of the connected two-point function of any fixed 0^{++} gauge-invariant operator varies weakly continuously in β on $[0.5, 2.5]$, with $\inf \text{supp}(d\rho_\beta) \geq m_0$ uniformly for some $m_0 > 0$.*

Then the mass gap $m(\beta) = \inf \text{supp}(d\rho_\beta)$ extends to a function holomorphic on the complex strip $S_{\varepsilon_0} = \{\beta \in \mathbb{C} : |\text{Im } \beta| < \varepsilon_0\}$ around the real interval $[0.5, 2.5]$, and in particular is real-analytic on the real interval itself.

Proof sketch (cf. [36]). By Osterwalder–Schrader reflection positivity [6] and the Glimm–Jaffe spectral construction [7, §19], the Wilson-loop expectation $\langle W(C) \rangle_{\Lambda, \beta}$ is meromorphic in β for each Λ , with poles only at the zeros of $Z_\Lambda(\beta)$. By (LY), there are no zeros in S_{ε_0} for any Λ_n . The family $\{\langle W(C) \rangle_{\Lambda_n, \beta}\}_n$ is uniformly bounded by 1 on S_{ε_0} (Wilson-loop magnitude bound for compact gauge group). Vitali’s convergence theorem [8, Thm. 14.6] gives that any subsequential thermodynamic limit is holomorphic on S_{ε_0} . The Bernstein–Hausdorff theorem [9] represents each fixed- β two-point function as a Stieltjes transform of the spectral measure $d\rho_\beta$, holomorphic in distance $|x|$ on the right half-plane. The abscissa of convergence of this Stieltjes family is $m(\beta)$, and by (SC) the family is holomorphic-jointly in β and $|x|$. The Kato perturbation theory of isolated eigenvalues [10, Thm. VII.1.7] transfers this to holomorphy of $m(\beta)$ on S_{ε_0} . $\square$

Corollary 5. *Conditional on Lemma 4 and on the existence of the YM measure in the thermodynamic limit on $[0.5, 2.5]$ as a constructive QFT object (an assumption strictly weaker than the W2-wall closure), positivity of $m(\beta)$ at the two endpoints $\beta = 0.5$ (CNS25 cluster expansion regime, extrapolated slightly past the rigorous frontier) and $\beta = 2.5$ (BCSK24 surface-sum regime, similarly extrapolated) together with absence of real-axis zeros of the analytic continuation, imply $m(\beta) > 0$ for every $\beta \in [0.5, 2.5]$.*

The corollary is the structural target. This paper proposes the measurement of (LY) for $SU(2)$. The companion programme [36] §4 articulates the test of (SC) via the FRG smoothness falsifiable F2.

3 Proposed computation

3.1 Modified Wang–Landau density of states

The Langfeld–Lucini–Rago algorithm [19] reconstructs the density of states $\rho_\Lambda(E)$ defined by the microcanonical decomposition (3), to high relative precision, via an iterative restricted-importance-sampling Monte Carlo. The key idea is to compute, for each energy interval $[E_i, E_{i+1}]$ partitioning the support $[-6, 6]$ of E , the ratio

$$\frac{\rho_\Lambda(E_{i+1})}{\rho_\Lambda(E_i)} = \exp(a_i \Delta E_i), \quad (5)$$

where a_i is a local linear coefficient (“LLR slope”) determined by a doubly truncated Gaussian importance sampling in the restricted energy window $[E_i - \delta, E_{i+1} + \delta]$. The slopes $\{a_i\}$ are accumulated iteratively (with Robbins–Monro stochastic approximation) and concatenated to give $\log \rho_\Lambda(E)$ up to an irrelevant additive constant. The relative precision achievable scales as $\sim 1/\sqrt{N_{\text{stat}}}$ per window in the LLR stochastic approximation regime, where N_{stat} is the number of Monte Carlo updates per window; with modern A100 GPU throughput $N_{\text{stat}} \sim 10^8$ – 10^{10} in single-digit days, comfortably yielding 60–80 digit relative precision on $\log \rho_\Lambda$ at $V = 4^4$ and 40–60 digits at $V = 6^4$ after standard accumulation across the ~ 240 windows spanning the $[-6, 6]$ energy range.

The validation of the modified Wang–Landau approach for $SU(2)$ specifically is contained in [19]. The relative precision required for our Lee–Yang application (see Section 3.3) is at least 40 digits to resolve $\varepsilon \sim 0.05$ – 0.5 at $V = 4^4$; 60 digits at $V = 6^4$; 80 digits at $V = 8^4$.

Table 1: Computational budget per lattice volume. “LLR steps” is the number of restricted importance-sampling updates per window; “Windows” is the number of energy intervals spanning $E \in [-6, 6]$ at width $\Delta E = 0.05$; “GPU days A100” is the estimated wall-clock cost on a single NVIDIA A100; “Digits” is the resulting relative precision on $\log \rho$.

V	LLR steps/window	Windows	GPU days A100	Digits
$4^4 = 256$	5×10^9	240	1–2	60–80
$6^4 = 1296$	2×10^9	240	5–10	40–60
$8^4 = 4096$	5×10^8	240	20–40	25–40

3.2 Complex- β partition function from real DoS

Given the real-axis DoS $\rho_\Lambda(E)$ reconstructed in Section 3.1, the complex- β partition function follows by analytic continuation of (3):

$$Z_\Lambda(\beta_0 + it) = \int_{-6}^6 \rho_\Lambda(E) e^{-(\beta_0 + it)VE/N} dE = e^{-\beta_0 V \langle E \rangle / N} \int_{-6}^6 \rho_\Lambda(E) e^{-(\beta_0 - \beta_0 + it)VE/N} dE \quad (6)$$

(after a trivial real-axis rescaling for numerical stability). The integrand is a real positive measure times an oscillating exponential, so (6) is precisely the *characteristic function* of the energy measure at imaginary “Fourier variable” tV/N .

Numerical implementation: *quadrature in arbitrary precision*. Standard adaptive Gauss–Legendre quadrature on a fine grid ($\sim 10^3$ – 10^4 knots) in E , in `arb` [38] or `mpmath` [39] or `PARI/GP` [37], performs the integration to the relative precision provided by the DoS. The zero-extraction prescription is then standard.

3.3 Cauchy zero-counting

To detect Lee–Yang zeros of $Z_\Lambda(\beta_0 + it)$ in the upper half-plane at fixed real β_0 , we use the standard Cauchy argument principle:

$$N_{\text{zeros}}(R) = \frac{1}{2\pi i} \oint_{C_R} \frac{Z'_\Lambda(\beta)}{Z_\Lambda(\beta)} d\beta, \quad (7)$$

where C_R is the rectangular contour $\{\beta_0 + it : 0 \leq t \leq R\} \cup \{\beta_0 + \delta + it : 0 \leq t \leq R\}$ (with reflections), closed by horizontal segments. By the symmetry $Z_\Lambda(\beta_0 - it) = \overline{Z_\Lambda(\beta_0 + it)}$, zeros in the lower half-plane mirror those in the upper, so the upper half is sufficient. The strip width $\varepsilon(\beta_0, V)$ is then the smallest R for which $N_{\text{zeros}}(R) \geq 1$ within the relevant grid resolution in $\text{Re } \beta$.

In practice we discretise the upper half-strip by a 2D grid $(\beta_0 + j\Delta\beta, k\Delta t)$ for $j \in \{-10, \dots, +10\}$ and $k \in \{1, \dots, K\}$, evaluate Z_Λ at each grid point via (6), and locate zeros by sign changes of $\text{Re } Z_\Lambda$ and $\text{Im } Z_\Lambda$ simultaneously (a square-grid Aberth-style refinement is then applied to localise each zero to 10^{-30}). The output is, for each real β_0 , the location of the nearest complex zero in the upper half-plane and hence $\varepsilon(\beta_0, V)$.

3.4 Lattice volumes and CPU budget

Table 1 summarises the per-volume computational budget. The total is $\lesssim 60$ A100-days, well within a single $\sim \$3000$ Vast.AI spend ($\$1.50$ – $\$2.50/\text{h}$ spot pricing for A100 80GB).

For $\text{SU}(3)$, the same protocol at $V = 4^4$ costs $\sim 3\times$ the $\text{SU}(2)$ figures (proportional to colour-group dimension and link variable size), still $\lesssim 10$ A100-days. We defer the $\text{SU}(3)$ computation to a Phase II programme; see Section 8.

4 Predictions

Prediction 6 (Finite strip width). For SU(2) lattice gauge theory in 4D, the Lee–Yang strip width $\varepsilon(\beta_0, V)$ admits a thermodynamic limit

$$\varepsilon_\infty(\beta_0) := \lim_{V \rightarrow \infty} \varepsilon(\beta_0, V) > 0 \quad \forall \beta_0 \in [0.5, 2.5]. \quad (8)$$

Concrete extrapolation prescription: $\varepsilon(\beta_0, V) = \varepsilon_\infty(\beta_0) + A(\beta_0) V^{-1/2} + O(V^{-1})$ with $A(\beta_0) > 0$, consistent with sub-Ising-like volume scaling in the absence of a transition.

Prediction 7 (Monotone non-increase across the crossover). The function $\beta_0 \mapsto \varepsilon_\infty(\beta_0)$ is non-increasing on $[0.5, 2.5]$, with

$$\varepsilon_\infty(2.5) \leq \varepsilon_\infty(0.5). \quad (9)$$

The expected ordering is consistent with the well-known smooth crossover from strong-coupling area-law (large ε , strong analyticity) to weak-coupling asymptotic-scaling regime (narrowing ε , weaker analyticity).

Prediction 8 (No pinching of the real axis). The infimum $\inf_{\beta_0 \in [0.5, 2.5]} \varepsilon_\infty(\beta_0)$ is strictly positive:

$$\varepsilon_0 := \inf_{\beta_0 \in [0.5, 2.5]} \varepsilon_\infty(\beta_0) \geq \varepsilon_0^{\min} > 0, \quad (10)$$

where $\varepsilon_0^{\min}$ is to be determined empirically. We predict $\varepsilon_0^{\min} \gtrsim 0.05$ in Wilson units on the basis of the Fröhlich–Spencer abelian analysis [4] extrapolated to non-abelian 4D, the FRG-extracted smoothness of the gluon propagator in [20], and the AT 2021 lattice data [12].

4.1 Falsification criteria

- Prediction 6 is *falsified* if, for any $\beta_0 \in [0.5, 2.5]$, the volume extrapolation gives $\varepsilon_\infty(\beta_0) = 0$ within 1σ statistical error at $V = 4^4, 6^4, 8^4$. This would indicate a real-axis phase transition at β_0 and would collapse Lemma 4 together with Corollary 5 at that β_0 .
- Prediction 7 is *falsified* if ε_∞ exhibits a strict interior maximum or any non-monotone feature beyond statistical noise. This would indicate non-trivial spectral substructure across the crossover (e.g. a higher-order analytic feature, perhaps a Lee–Yang “edge singularity” analogue of the Yang–Lee universality class).
- Prediction 8 is *falsified* if $\varepsilon_0 < 0.01$ in Wilson units at any tested β_0 . Such a small strip would imply borderline analyticity (the synthesis lemma still holds, but the analytic continuation domain is too narrow for downstream applications like Padé–Borel summation of the FRG flow).

5 Expected results

5.1 Order-of-magnitude estimate from FRG smoothness

The functional renormalisation group computation of [20] for pure SU(3) Yang–Mills in Landau gauge produces a gluon propagator $D_g(p^2; \beta)$ that is smooth in β to all derivative orders tested. By analogy with abelian lattice Coulomb gas analyticity bounds [4, 5], smoothness in β to derivative order k implies a Lee–Yang strip width $\varepsilon \gtrsim k^{-1}$ in standard normalisations. The FRG data is smooth to $k \geq 8$ orders empirically; a $\sim O(1/8)$ strip width estimate gives $\varepsilon \sim 0.1$ in ’t Hooft units, which for SU(2) at $\beta_W = 2$ corresponds to ~ 0.05 in Wilson units. This matches the order of magnitude of Prediction 8.

Table 2: Expected order-of-magnitude $\varepsilon_\infty(\beta_0)$ in Wilson units for SU(2). Values for $\beta = 0.1, 2.5$ are *lower bounds* from CNS25 and BCSK24 rigorous disks; the interior values are predictions of this paper.

β_W	0.1	0.5	1.0	1.5	2.0	2.5	5
λ_t	80	16	8	5.3	4	3.2	1
ε_∞ (predicted)	≥ 0.33	0.3	0.15	0.08	0.06	≥ 1.5	≥ 1
Regime	CNS25 rigorous	CNS frontier	W2 wall	W2 wall	W2 wall	BCSK frontier	BCSK24

5.2 Order-of-magnitude estimate from AT 2021

The 8-anchor SU(2.8) scan of [12] shows no first-order discontinuity in the 0^{++} glueball mass at any β on [2.3, 2.7] to $\sim 5\%$ precision. The corresponding absence of Lee–Yang real-axis zeros is consistent with $\varepsilon \gtrsim (5\% \text{ lattice precision}) \times L^{-1/2} \sim 0.05$ at $L = 24$, again of order 0.1 in conservative Wilson units. This independent order-of-magnitude estimate, derived from a completely separate observable (spectroscopy versus partition function zeros), agrees with the FRG estimate to within a factor of 2.

5.3 Endpoint comparison: CNS25 and BCSK24

At the strong-coupling endpoint $\beta_W \rightarrow 0$, the Cao–Nissim–Sheffield cluster expansion [25, 26] provides explicit analyticity on a disk of radius r_{CNS} that, in Wilson units, satisfies $r_{\text{CNS}} > \frac{1}{3}$ for SU(2). Direct computation in our DoS reconstruction at $\beta_W = 0.1$ should reproduce $\varepsilon(\beta_W = 0.1, \infty) \gtrsim r_{\text{CNS}} > \frac{1}{3}$, providing a *cross-check* of the DoS pipeline against rigorous analytic results.

Similarly, at the weak-coupling endpoint $\beta_W = 2.5$ the Borga–Cao–Shogren–Knaak surface sum [27] provides analyticity on a disk of radius r_{BCSK} with $r_{\text{BCSK}} > 1.5$ in Wilson units for SU(2), so $\varepsilon(\beta_W = 2.5, \infty) \gtrsim 1.5$. The interesting question is the interior: we predict that ε takes its minimum somewhere on [0.5, 1.0] (in ’t Hooft units, near $\lambda_t \sim 6\text{--}8$), where the smooth crossover between the two regimes is most active.

5.4 Summary table of expected outcomes

Table 2 summarises the expected shape of $\varepsilon_\infty(\beta_0)$. The salient feature is the “trough” in the middle of the W2 wall, near $\beta_W \in [1.0, 2.0]$, where the strip is narrowest. Even at the trough, we predict $\varepsilon \gtrsim 0.05$, comfortably above the falsification threshold of Prediction 8.

6 ROI analysis

Table 3 summarises the cost-benefit analysis.

Prior probability of PASS. We assess $P(\text{PASS}) \approx 80\%$ on the basis of:

1. Empirical absence of a real-axis transition in $\beta_W \in [0.5, 2.5]$ across all published SU(2) lattice data since [13] (qualitative match ~ 25 papers).
2. Smoothness of the FRG-extracted gluon propagator and glueball mass across the same interval to derivative order ≥ 8 [20, 23].
3. Endpoint rigorous control via CNS25 [25, 26] and BCSK24 [27] on overlapping (in ’t Hooft units) disks; these alone exclude the strong-/weak-coupling pinching.
4. Theoretical Fröhlich–Spencer [4] template for abelian 4D gauge: zero-free strip persists across the crossover; non-abelian generalisation is structurally expected.

Table 3: Cost-benefit analysis for the proposed Lee–Yang strip measurement for $SU(2)$. “Probability(PASS)” is our prior probability of a positive outcome; “Uplift if PASS” is the expected gain in the W2-wall analyticity closure probability on the 10-year horizon, conditional on PASS.

Item	Value
GPU time on single A100 ($SU(2)$, all V)	30–60 days
Vast.AI cost (\$1.50–\$2.50/h spot)	~\$1500–\$3500
Human time (analysis + writeup)	1–2 person-months
Total estimated cost	~\$3000 + person-months
Prior $P(\text{PASS}) = P(\varepsilon_0 > 0)$	~80%
Uplift to W2-wall closure P_{10y} if PASS	+25–30 pp
Updated W2 closure P_{10y} if PASS	~50%
Updated W2 closure P_{10y} if FAIL	~10% (with new identified obstruction)
Publishability if PASS	TIER SUPPORTED of synthesis
Publishability if FAIL	TIER FALSIFIED with new obstruction
ROI (unitless)	≈ 8

We assess $P(\text{FAIL}) \approx 20\%$, where the failure mode would be a finite-volume Lee–Yang zero pinching as V grows at some interior β_0 . Even in this case, the result is publishable as a TIER PROVED NEGATIVE for the analyticity synthesis: such a pinch would constitute a *specific obstruction* (a complex-coupling phase transition not seen in the real spectroscopy), pointing at the failure of sub-claim (LY) in a quantitative way that opens new theoretical questions about the spectral structure across the wall.

Uplift to W2-wall closure probability. Conditional on PASS, sub-claim (LY) of Lemma 4 is TIER SUPPORTED-empirical. The only remaining open sub-claim is (SC), which is the subject of Falsifiable F2 of [36] §4.2 (FRG smoothness density test). Given the strong theoretical and empirical evidence already available for (SC), conditional probability $P(\text{SC} | \text{LY})$ is ~60%, so a successful F1 brings W2-wall analyticity (Lemma 4 hypotheses fully met) to ~ 0.5 on the 10-year horizon, up from the ~22% union-bound prior [36] §2.4. This is the headline numerical claim of the ROI analysis.

7 Implications for the W2 wall

7.1 The synthesis status update if PASS

If the F1 computation returns PASS (Predictions 6–8 confirmed empirically), then:

1. Sub-claim (LY) of Lemma 4 is TIER SUPPORTED-empirical, with explicit numerical bounds $\varepsilon_0 \geq \varepsilon_0^{\min}$ for use in downstream constructive arguments.
2. The synthesis lemma is at TIER SKETCHED + CONDITIONAL ON (SC); the conditional is the single remaining obstruction and is itself falsifiable in a separate computation [36, §4.2].
3. Padé–Borel resummation of the FRG flow [20] on the strip S_{ε_0} becomes *rigorous* (the analytic-continuation domain exists), upgrading the FRG mass-gap extraction from a self-consistent truncation to a controlled asymptotic expansion.
4. The corpus of 165 exact cross-Galois Q -rationals at $h_K \in \{1, 2, 4, 8, 16\}$ documented in [35] §M.B can be re-interpreted as Padé interpolation anchors for $m(\beta, N)$ via the $F(N)$

parameter-free [35, Theorem D'] §M.A.1 factorisation, on a now-rigorous holomorphic strip¹.

5. The headline W2-wall closure probability on the 10-year horizon moves from $\sim 22\%$ to $\sim 50\%$ (cf. ROI table).

7.2 The synthesis status update if FAIL

If F1 returns FAIL at some interior $\beta_0 \in [0.5, 2.5]$, then:

1. Sub-claim (LY) is TIER FALSIFIED; the synthesis lemma collapses at that β_0 .
2. However, the failure mode is *informative*: a Lee–Yang zero pinching the real axis at β_0 corresponds to a *complex-coupling phase transition* not visible in real observables. This is a major structural discovery, opening new questions: which spectral mode becomes singular at the complex transition? Is it the (gauge-variant) gluon two-point function, or a (gauge-invariant) glueball channel?
3. Publishability is unchanged: a paper announcing TIER PROVED NEGATIVE “Lee–Yang zeros of SU(2) lattice YM pinch the real axis at $\beta_0 = \beta_\star$ ” is a significant result for the constructive QFT community, even (or especially) given the empirical absence of a real spectroscopy transition.
4. The W2-wall closure probability on the 10-year horizon drops to $\sim 10\%$, but with an identified specific obstruction (target for the next 5–10 years of theoretical work).

7.3 Either way, the F1 computation is decisive

The cardinal feature of the proposed programme is its *epistemic decisiveness*: with probability $\geq 98\%$, the result distinguishes between two qualitatively different descriptions of the W2 wall (Lee–Yang strip survives versus collapses). This is the highest-resolution single experiment that can be performed on the W2-wall analyticity question at current cost, and we recommend its execution as the top-ROI deliverable of the next 6 months.

8 Open questions

8.1 Extension to SU(3) and SU(N)

The same protocol applied to SU(3) at $V = 4^4$ costs $\sim 3\times$ the SU(2) figures of Table 1, still $\lesssim 10$ A100-days; the SU(3) computation is therefore tractable in a Phase II programme at $\sim \$5000$ – $\$8000$ additional cost. The $SU(N)$, $N \geq 4$ extension is exponentially expensive in N due to the link variable dimension, but the 't Hooft scaling predicts an *N-independent* strip width $\varepsilon(\lambda_t)$ in 't Hooft units; testing this prediction at SU(2), SU(3) (and tentatively SU(4)) constitutes the natural follow-up.

8.2 Connection to instanton density

The topological-charge density $q(x)$ on the lattice is closely related to the imaginary part of the gauge action in the complex extension: $\text{Im } S(U)|_{\text{Im } \beta \neq 0} \sim \theta(\beta) \cdot Q_{\text{top}}$, with $\theta(\beta)$ the β -dependent effective theta-angle. The Lee–Yang zeros at $\text{Im } \beta = \varepsilon$ correspond to constructive interference of topological-charge sectors, and the strip width $\varepsilon(\beta_0)$ should be *anti-correlated*

¹Conditional on the additional F(N) factorisation sub-claim, itself a falsifiable target (F3 of [36] §4.3) at a cost of 6–12 person-months.

with the topological susceptibility $\chi_t(\beta_0)$. This is testable: at the BDLV [18] freeze-resolved SU(3) anchor $\beta_W \approx 6.3$, $\chi_t^{1/4} \approx 191$ MeV gives a quantitative prediction for ε that can be cross-checked against the F1 computation Phase II.

8.3 Relation to Fröhlich–Spencer abelian Lee–Yang

The Fröhlich–Spencer 1982 result [4] establishes a Lee–Yang strip for compact U(1) lattice gauge in 4D that vanishes at the deconfinement transition. The non-abelian generalisation we propose has no real-axis transition (smooth crossover), so the strip is predicted to persist; but the microscopic mechanism by which the U(1) strip shrinks (Coulomb-gas energy-entropy balance) is not directly transferable to SU(2) (no monopole sector in the same sense, but vortex sector [31] present). A clean understanding of why the U(1) strip collapses while the SU(2) strip persists, even in the absence of confinement-deconfinement transitions in 4D SU(2), would be a foundational result of independent interest.

8.4 Spectral measure continuity (sub-claim SC)

This paper addresses only sub-claim (LY). The companion falsifiable F2 [36, §4.2] is the test of (SC): repeat the Cyrol et al. 2016 [20] computation at 50 β values densely sampling [0.5, 2.5] and numerically differentiate the extracted mass to order $k \leq 10$. A successful F2 (no kink or divergence in any derivative) together with the F1 result of the present paper constitute the complete empirical support for the synthesis Lemma 4 hypotheses. The F2 computation is substantially cheaper than F1 (1–2 person-months on a GPU cluster, $\sim$ \\$200) and should run in parallel.

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Cluster firm 404 STABLE entry and exit; no new arXiv identifiers introduced. Every arXiv ID cited below appears in the ECI v15 BIGTABLE living catalog [35] §M.G with explicit verification mark, and was WebFetch-confirmed in the predecessor analyticity report [36].

A Pseudocode for the Lee–Yang strip measurement

The following pseudocode summarises the complete F1 pipeline. The implementation language is intentionally left abstract; in production we anticipate a mix of C (LLR Monte Carlo on GPU) + PARI/GP (arbitrary-precision integration and zero extraction).

```

INPUT: lattice volume V; gauge group SU(N); coupling grid
      beta0_grid = [0.5, 0.6, ..., 2.5]; imaginary-coupling
      grid t_grid = [1e-4, 2e-4, ..., 2.0]; precision PREC digits.

% PHASE 1: Density of states (Langfeld-Lucini-Rago modified Wang-Landau)
FOR each energy window [E_i, E_{i+1}], i = 1, ..., W:
```

```

    initialise gauge configuration U with mean plaquette in window
    initialise LLR slope a_i = 0
    FOR k = 1, ..., N_iter (Robbins-Monro stochastic approximation):
        run N_MC restricted-window Metropolis updates of U
        sample <E - E_mid> conditional on E in [E_i, E_{i+1}]
        update a_i <- a_i + alpha_k * <E - E_mid> (alpha_k = 1/k)
    record a_i
END FOR
reconstruct log(rho_Lambda(E)) by concatenation:
    log(rho(E_{i+1})) - log(rho(E_i)) = a_i * (E_{i+1} - E_i)

% PHASE 2: Complex-beta partition function from DoS
FOR each (beta0, t) in beta0_grid x t_grid:
    Z(beta0 + i*t) = integrate over E from -6 to 6 of
        rho(E) * exp(-(beta0 + i*t)*V*E/N) dE
    use arbitrary-precision Gauss-Legendre quadrature, PREC digits
    record Z_real, Z_imag at each grid point
END FOR

% PHASE 3: Lee-Yang zero extraction via Cauchy + Aberth refinement
FOR each beta0 in beta0_grid:
    locate sign changes of Z_real and Z_imag simultaneously in t_grid
    refine each candidate zero via Aberth iteration to 1e-30 precision
    epsilon(beta0, V) = min{ t : Z(beta0 + i*t) = 0 }
END FOR

% PHASE 4: Volume extrapolation
FOR each beta0 in beta0_grid:
    fit epsilon(beta0, V) = epsilon_inf(beta0) + A(beta0)*V^{-1/2}
    extract epsilon_inf(beta0) and uncertainty
END FOR

OUTPUT: epsilon_inf(beta0) for beta0 in [0.5, 2.5];
        verdict: PASS (epsilon_inf > 0.01 everywhere) or
                FAIL (epsilon_inf < 0.01 at some beta0).

```

The same pipeline applies to $SU(3)$ with the natural modifications: the link variable is $SU(3)$ -valued, the plaquette action range is $[-6V \cdot 3, 6V \cdot 3]$, and the per-window MC cost is $\sim 3 \times$ the $SU(2)$ cost (proportional to colour-group dimension).

B Convention summary

For ease of reference and cross-paper consistency:

- *Wilson normalisation*: $\beta_W = 2N/g^2$, action $S = (\beta_W/N) \sum_p \text{Re Tr } U_p$.
- *'t Hooft normalisation*: $\lambda_t = 2N^2/\beta_W$; equivalently $\lambda_t = g^2 N$.
- $SU(2)$: $\beta_W = 8/\lambda_t$, range $\lambda_t \in (2, 24) \Leftrightarrow \beta_W \in (1/3, 4)$.
- $SU(3)$: $\beta_W = 18/\lambda_t$, range $\lambda_t \in (2, 24) \Leftrightarrow \beta_W \in (3/4, 9)$.
- Mission interval $\beta_W \in [0.5, 2.5]$ for $SU(2)$ corresponds to $\lambda_t \in [3.2, 16]$, entirely inside the W2 wall.

- Physical lattice spectroscopy regime for $SU(2)$: $\beta_W \in [2.3, 2.7]$ (AT 2021 [12, §5]), i.e. near the BCSK frontier in Wilson units but well past it in 'tHooft units.

References

- [1] T. D. Lee and C. N. Yang, *Statistical theory of equations of state and phase transitions. II. Lattice gas and Ising model*, Phys. Rev. **87** (1952) 410–419.
- [2] M. Suzuki and M. E. Fisher, *Zeros of the partition function for the Heisenberg, ferroelectric, and general Ising models*, J. Math. Phys. **12** (1971) 235–246.
- [3] D. Ruelle, *Statistical Mechanics: Rigorous Results*, W. A. Benjamin (1969).
- [4] J. Fröhlich and T. Spencer, *Massless phases and symmetry restoration in abelian gauge theories and spin systems*, Commun. Math. Phys. **83** (1982) 411–454.
- [5] D. Brydges and P. Federbush, *Debye screening*, Commun. Math. Phys. **73** (1980) 197–246.
- [6] K. Osterwalder and R. Schrader, *Axioms for Euclidean Green's functions*, Commun. Math. Phys. **31** (1973) 83–112.
- [7] J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, Springer (1981).
- [8] E. Hille, *Analytic Function Theory*, vol. II, Chelsea (1962).
- [9] S. Bernstein, *Sur les fonctions absolument monotones*, Acta Math. **52** (1928) 1–66.
- [10] T. Kato, *Perturbation Theory for Linear Operators*, Springer (1995), reprint of the 1980 edition.
- [11] A. Jaffe and E. Witten, *Quantum Yang–Mills theory*, Clay Mathematics Institute Millennium Problem description (2000).
- [12] A. Athenodorou and M. Teper, *$SU(N)$ gauge theories in 3+1 dimensions: glueball spectrum, string tensions and topology*, JHEP **12** (2021) 082, [arXiv:2106.00364](#).
- [13] B. Lucini and M. Teper, *$SU(N)$ gauge theories in four dimensions: exploring the approach to $N = \infty$* , JHEP **06** (2001) 050, [arXiv:hep-lat/0103027](#).
- [14] B. Lucini, M. Teper and U. Wenger, *Glueballs and k -strings in $SU(N)$ gauge theories: calculations with improved operators*, JHEP **06** (2004) 012, [arXiv:hep-lat/0404008](#).
- [15] S. Necco and R. Sommer, *The $N_f = 0$ heavy quark potential from short to intermediate distances*, Nucl. Phys. B **622** (2002) 328–346, [arXiv:hep-lat/0108008](#).
- [16] B. Lucini, A. Rago and E. Rinaldi, *Glueball masses in the large N limit*, JHEP **08** (2010) 119, [arXiv:1007.3879](#).
- [17] B. Lucini and M. Panero, *$SU(N)$ gauge theories at large N* , Phys. Rept. **526** (2013) 93–163, [arXiv:1210.4997](#).
- [18] C. Bonanno, M. D’Elia, B. Lucini and D. Vadicchino, *Towards glueball masses of large- N $SU(N)$ pure-gauge theories without topological freezing*, Phys. Lett. B **833** (2022) 137281, [arXiv:2205.06190](#).

- [19] K. Langfeld, B. Lucini and A. Rago, *The density of states in gauge theories*, Phys. Rev. Lett. **109** (2012) 111601, [arXiv:1204.3243](#).
- [20] A. K. Cyrol, L. Fister, M. Mitter, J. M. Pawłowski and N. Strodthoff, *Landau gauge Yang–Mills correlation functions*, Phys. Rev. D **94** (2016) 054005, [arXiv:1605.01856](#).
- [21] A. K. Cyrol, M. Mitter, J. M. Pawłowski and N. Strodthoff, *Non-perturbative quark, gluon and meson correlators of unquenched QCD*, Phys. Rev. D **97** (2018) 054006, [arXiv:1706.06326](#).
- [22] C. Wetterich, *Exact evolution equation for the effective potential*, Phys. Lett. B **301** (1993) 90–94, [arXiv:1710.05815](#).
- [23] M. Q. Huber, C. S. Fischer and H. Sanchis-Alepuz, *Apparent convergence in functional glueball calculations*, 2025, [arXiv:2503.03821](#).
- [24] H. Sanchis-Alepuz, C. S. Fischer, C. Kellermann and L. von Smekal, *Glueballs from the Bethe–Salpeter equation*, Phys. Rev. D **92** (2015) 034001, [arXiv:1503.06051](#).
- [25] S. Cao, A. Nissim and S. Sheffield, *Expanded regimes of area law for 4D lattice Yang–Mills*, 2025, [arXiv:2505.16585](#).
- [26] S. Cao, A. Nissim and S. Sheffield, *Area law for lattice Yang–Mills*, 2025, [arXiv:2509.04688](#).
- [27] M. Borga, S. Cao, J. Shogren and A. Knaak, *Surface sums for lattice Yang–Mills in the large- N regime*, 2024, [arXiv:2411.11676](#).
- [28] S. Cao, M. Park and S. Sheffield, *Random surfaces and lattice Yang–Mills*, 2023, [arXiv:2307.06790](#), accepted Commun. Amer. Math. Soc.
- [29] A. Nissim, *$U(N)$ lattice Yang–Mills in the ’t Hooft regime*, 2025, [arXiv:2510.22788](#).
- [30] G. ’t Hooft, *A planar diagram theory for strong interactions*, Nucl. Phys. B **72** (1974) 461–473.
- [31] G. ’t Hooft, *On the phase transition towards permanent quark confinement*, Nucl. Phys. B **138** (1978) 1–25.
- [32] M. Hairer, *A theory of regularity structures*, Invent. Math. **198** (2014) 269–504, [arXiv:1303.5113](#).
- [33] S. Cao, A. Chandra, I. Chevyrev, M. Hairer and H. Shen, *Stochastic quantisation of Yang–Mills in 2D*, 2020, [arXiv:2006.04987](#).
- [34] S. Chatterjee, *Yang–Mills correlations on the three-dimensional lattice*, 2024, [arXiv:2201.03487](#).
- [35] K. Rémondière, *ECI v15 living catalog (BIGTABLE v2)*, 17 May 2026, internal note `OP_BIGTABLE_V2_2026-05-17.md`.
- [36] K. Rémondière, *Analyticity of the YM mass gap $m(\beta)$ on $\beta \in [0.5, 2.5]$: three routes and the reflection-positivity + Stieltjes + Vitali synthesis*, 17 May 2026, internal note `OP_W2_ANALYTICITE_2026-05-17.md`.
- [37] The PARI Group, *PARI/GP version 2.15.4*, Univ. Bordeaux (2023), <http://pari.math.u-bordeaux.fr/>.

- [38] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017) 1281–1292.
- [39] F. Johansson et al., *mpmath: a Python library for arbitrary-precision floating-point arithmetic*, version 1.3.0 (2023), <http://mpmath.org/>.